

Feedback - Research Proposal Outline

Thank you for submitting your presentation outline, “Developing an AI-Optimized IoT Framework for Sustainable Water Management in the UAE.” This is a strong, well-structured proposal that is clearly aligned with MSc Artificial Intelligence expectations and the requirements of the Research Methods and Professional Practice module.

The significance and research problem are articulated very effectively. You clearly contextualise the UAE’s water scarcity challenge and convincingly justify why current IoT-based systems are insufficient without advanced AI-driven analytics. The link to national sustainability and smart city goals strengthens both academic and practical relevance.

Your research question, aim, and objectives are clear, focused, and ambitious but achievable. They demonstrate a good balance between conceptual contribution and artefact development. The emphasis on predictive analytics (leak detection, demand forecasting, anomaly detection) is particularly appropriate for an AI-focused MSc project.

The literature framing is sound, clearly identifying a gap between IoT data collection and intelligent decision-making. You appropriately position AI techniques as the core contribution of your study. To strengthen this further, ensure your final presentation critically contrasts existing IoT-only solutions with AI-enhanced approaches rather than simply describing them.

The methodology is a major strength. Design Science Research is well suited to this project, and the phased approach from design to evaluation is clearly explained. You could enhance rigour by briefly clarifying how evaluation metrics will be selected and interpreted in relation to sustainability outcomes.

Your ethical considerations and risk assessment are thoughtful and realistic, especially regarding data bias, automation impacts, and scope control. The artefact description is clear, feasible, and appropriate for assessment.